

Conditional Structures & Control Flow

COMPREHENSIVE SOLUTIONS & STUDY GUIDE

1. What is a conditional statement in programming?

A conditional statement is a fundamental programming construct that allows a program to execute different blocks of code depending on whether a specified expression evaluates to `true` or `false`. It enables decision-making and controls the flow of execution within an application.

2. What does an if statement do?

An `if` statement evaluates a specified Boolean condition. If the condition evaluates to `true`, the block of code inside the statement is executed. If it evaluates to `false`, the code block is completely bypassed, and control shifts to the next sequential statement outside the block.

3. What is the purpose of an else block?

The purpose of an `else` block is to define an alternative path of execution. It guarantees that a specific block of code will run if and only if the primary condition in the preceding `if` statement evaluates to `false`.

4. When is an else block executed?

An `else` block is executed when all preceding conditions in the conditional chain (the initial `if` statement and any intermediate `else if` statements) have evaluated to `false`.

5. What is a Boolean expression?

A Boolean expression is an algebraic or logical expression that produces a value belonging to a binary truth domain—yielding exactly one of two possible outputs: `true` or `false`.

6. Write the syntax of a simple if statement.

```
if (condition) {  
    // Code to execute if the condition evaluates to true  
}
```

7. What operator is commonly used to test equality?

The double equals operator (`==`) is commonly used to test equality between two operands. It is distinct from the single equals sign (`=`), which is the assignment operator.

8. What does the modulus (%) operator return?

The modulus operator (`%`) returns the remainder resulting from an integer division operation between two numbers. For example, `5 % 2` returns `1`.

9. How can you check if a number is even?

You can check if a number is even by evaluating whether its remainder when divided by 2 is exactly zero using the modulus operator: `if (number % 2 == 0)`.

10. How can you check if a number is odd?

You can check if a number is odd by evaluating whether its remainder when divided by 2 is not equal to zero: `if (number % 2 != 0)` or `if (number % 2 == 1)`.

11. What is the purpose of else-if?

The `else if` clause allows you to test multiple sequential and mutually exclusive conditions. If the initial `if` statement is false, the program proceeds down the line to check subsequent `else if` expressions.

12. What is a nested if statement?

A nested `if` statement is an `if` statement that is completely enclosed inside the body of another `if` or `else` block. It is used to evaluate hierarchical or multi-stage conditions.

13. Can an if statement exist without an else?

Yes, an `if` statement can exist entirely on its own. An `else` block is completely optional and is omitted when no action is required when the condition is false.

14. What is the output when a false condition is tested in an if statement without else?

There is no output or execution from that block. The program simply skips over the conditional block and continues executing the statements that follow it.

15. What is the role of braces {} in C/C++ conditionals?

Curly braces `{ }` define a compound block of statements. They group multiple instructions together so they are treated as a single structural unit controlled by the conditional statement. If omitted, only the single immediately following statement falls under the scope of the conditional.

16. What does > mean?

It is the relational **greater than** operator. It evaluates to `true` if the left operand is strictly larger than the right operand.

17. What does < mean?

It is the relational **less than** operator. It evaluates to `true` if the left operand is strictly smaller than the right operand.

18. What does >= mean?

It is the relational **greater than or equal to** operator. It evaluates to `true` if the left operand is either larger than or exactly equal to the right operand.

19. What does <= mean?

It is the relational **less than or equal to** operator. It evaluates to `true` if the left operand is either smaller than or exactly equal to the right operand.

20. What does != mean?

It is the relational **not equal to** operator. It evaluates to `true` if the two operands being compared do not possess the same value.

21. What is a switch statement?

A `switch` statement is a control flow structure that provides an elegant alternative to long `if-else-if` ladders. It evaluates an expression once and branches execution to a matching `case` label based on constant integral values.

22. What is a case label in a switch statement?

A case label (e.g., `case 5:`) specifies a constant value within a `switch` structure. If the control expression matches this value, execution jumps directly to the statements following this label.

23. Why is break used in switch?

The `break` statement is used to terminate execution within the `switch` block immediately. It forces the program to jump out of the switch structure, preventing execution from bleeding into subsequent cases.

24. What happens if break is omitted?

If `break` is omitted, execution falls through sequentially into the next case statements below it, executing their associated code even if those case labels do not match the control expression.

25. What is the default case in switch?

The `default` case is an optional catch-all block inside a `switch` statement. It executes only if none of the explicit `case` values match the evaluated control expression.

26. Can switch compare strings in C++?

No, traditional `switch` statements in C++ can only evaluate variables that resolve to integral types (such as `int`, `char`, or `enum`). They do not directly support object types like strings.

27. Which statement is better for many constant choices: if-else or switch?

A `switch` statement is generally superior for checking a single variable against many constant integral choices. It is highly readable, less prone to formatting errors, and can be optimized by compilers via lookup jump tables.

28. How do you test whether a value is positive?

You check if the value is strictly greater than zero: `if (value > 0)`.

29. How do you test whether a value is negative?

You check if the value is strictly less than zero: `if (value < 0)`.

30. How do you test whether a value is zero?

You check for equality with zero: `if (value == 0)`.

31. Write a condition to check if age is at least 18.

```
if (age >= 18)
```

32. Write a condition to check if marks are above 50.

```
if (marks > 50)
```

33. What is decision making in programming?

Decision making is the structural methodology where programmers write rules that cause the system to dynamically select an execution path at runtime based on the evaluation of conditional logic and state data.

34. Can nested if statements have multiple levels?

Yes, you can nest `if` statements across multiple levels deep. However, deeply nested structures should be minimized as they degrade code readability and maintainability (often referred to as the "arrow anti-pattern").

35. What is logical AND (&&)?

Logical AND (`&&`) is a binary operator that joins two Boolean expressions. It yields `true` if and only if both individual expressions are `true`. It exhibits short-circuit evaluation.

36. What is logical OR (||)?

Logical OR (`||`) is a binary operator that joins two Boolean expressions. It yields `true` if at least one of the expressions is evaluated as `true`.

37. What is logical NOT (!)?

Logical NOT (`!`) is a unary operator used to invert the truth value of its operand. It transforms `true` into `false`, and `false` into `true`.

38. How do you check if a year is divisible by 4?

You use the modulus operator to check if dividing the year by 4 leaves a remainder of zero: `if (year % 4 == 0)`.

39. What is a leap year?

A leap year is an astronomical correction year containing 366 days instead of 365. The extra day is added to February to realign the calendar year with the Earth's orbital period around the sun.

40. What is the first condition for a leap year?

The first baseline criteria is that the year must be evenly divisible by 4 (`year % 4 == 0`). Complete leap year rules also stipulate that century years must additionally be divisible by 400.

41. How do you find the largest of two numbers?

```
if (num1 > num2) {  
    largest = num1;  
} else {  
    largest = num2;  
}
```

42. How do you find the largest of three numbers?

```
if (a >= b && a >= c) {  
    largest = a;  
} else if (b >= a && b >= c) {  
    largest = b;  
} else {  
    largest = c;  
}
```

43. What is an if-else-if ladder?

An `if-else-if` ladder is a continuous chain of conditional tests. Expressions are checked from top to bottom. As soon as a condition evaluates to `true`, its code runs, and the entire remainder of the chain is skipped.

44. What is the difference between if and switch?

An `if` statement evaluates broad relational and logical compound conditional bounds (like ranges, strings, or complex expressions). A `switch` statement evaluates a single integral value against absolute, pre-defined constant labels.

45. Can switch use ranges directly?

No, standard ISO C/C++ does not natively support range evaluations directly inside case statements. However, some individual compilers (like GCC) support range syntax as a non-standard syntax extension (e.g., `case 1 ... 5:`).

46. What is fall-through in switch?

Fall-through is the behavior where execution continues sequentially into adjacent downward cases inside a `switch` block because a `break` command was not supplied to stop it.

47. What data type is usually used for conditions?

The **Boolean** data type (represented as `bool` in C++) is the standard type used for storing and evaluating logical true or false conditions.

48. Why are conditional statements important?

Without conditional statements, programs would be completely static and linear, running exactly the same steps every single time. Conditionals allow code to respond to dynamic input variations, handle interactive behavior, and execute sophisticated decision logic.

49. Give one real-life example of a conditional statement.

ATM Withdrawal: `IF` your requested withdrawal amount is less than or equal to your current account balance, `THEN` dispense the cash and deduct the funds; `ELSE`, display an "Insufficient Funds" error notice.

50. Name four conditional structures covered.

The four standard control flow structures are: the standalone `if` statement, the `if-else` statement block, the multi-tiered `if-else-if` ladder, and the multi-branch `switch` statement.